



KNOW HOW



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❶ I have just been introduced to **LINUX For You**. I was really surprised to see that a complete operating system was bundled free on the DVD (openSUSE 11.2). I want to know how to use that DVD to install the OS. Do I need to remove Windows XP from my system or will openSUSE work along with it?

—Rajiv Kumar,
rajiv4sap@yahoo.com

A. You can always install Linux along with Windows XP and configure your system with the dual-boot option. I would recommend that you take a complete back-up of your data before trying to install openSUSE 11.2 on your system – just to establish a good practice. Do also try to confirm that your system meets all requirements before trying out the install DVD. Boot your system using the DVD and follow the on-screen message to complete your installation.

The detailed installation guide is also provided on the DVD itself. Go to the “docu” folder on the root directory of the DVD and open the file `startup_en.pdf`.

❷ I have a small network with a few computers and a laptop.

A few days back, I migrated all systems to a Linux-based OS. I just want to know if there's any tool using which I can check what ports are open on the different computers.

—Pawan,
prasad.pawan@gmail.com

A. You need to install Nmap on your system. As mentioned in the man pages, “Nmap is an open source tool for network exploration and security auditing. It was designed to rapidly scan large networks, although it works fine against single hosts.” Basic usage of Nmap is simple. Just invoke it with a hostname, or IP address. To scan for the open ports, issue the following command:

```
#nmap <IP Address>
```

For example:

```
[root@test ~]# nmap 192.168.0.222
Starting Nmap 4.20 ( http://nmap.org ) at
2010-08-17 18:40 IST
Interesting ports on 192.168.0.222:
Not shown: 1695 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp    open  rpcbind
MAC Address: 08:22:01:AB:AA:00 (Unknown)
Nmap finished: 1 IP address (1 host up) scanned
in 1.119 seconds
```

You can specify `-p 100-125`; then Nmap will scan on ports 100 to 125 only, and will let you know the open and closed ports. Please go through the man pages for more details.

❸ I am a regular reader of **LFY** magazine. I have Fedora 13 installed on my system. I want to connect to the Net using a

Micromax MMX310G modem. I used “`wvdial`” but the modem could not be detected by the system. Could you please help me?

—Bhaskar Ghosh,
bhaskarghsh4@gmail.com

A. These new modems work in dual mode – one is the storage device mode where it works as a USB CD-ROM and the other is the modem mode. So, the first thing to check is whether your system recognises the device as a modem or not. Check the `dmesg` and confirm this. If it is recognised as a modem, then all you need to do is to run the following command:

```
#wvdialconf /etc/wvdial.conf
```

and fire

```
#wvdial
```

But if the USB device is recognised as the CD-ROM, then you need to switch it to USB modem mode.

To do this, you need to have USB_ModeSwitch, which is quite handy for experimenting with your own hardware if it's not supported yet.

Download the package from http://www.draisberghof.de/usb_modeswitch/usb-modeswitch-1.1.4.tar.bz2 and extract it. Go through the *Readme* file to learn how to install. Check `/proc/bus/usb/devices` or `lspci` to find out the product and vendor ID of the device. Read the `device_reference.txt` and do as instructed.

Once you are able to switch to USB mode, run the following command:

```
#modprobe usbserial vendor=<vendor_id>
product=<product_id>
```

Now run `wvdialconf` and configure your modem. 